

Initiating Search

November 11, 2025, 5:17 PM

• Search:

CAS Lexicon Search:

Nitriles - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (25,977)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,631)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

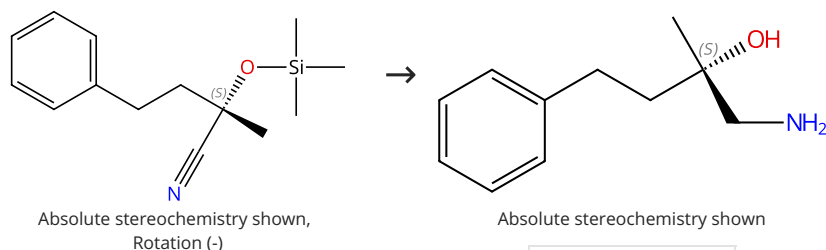
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

[Suppliers \(4\)](#)

31-614-CAS-32013682

Steps: 1 Yield: 100%

Organocatalytic stereoselective cyanosilylation of small ketones

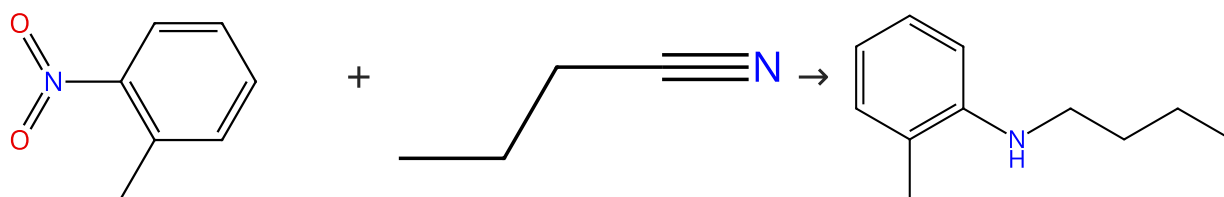
By: Zhou, Hui; et al

Nature (London, United Kingdom) (2022), 605(7908), 84-89.

- 1.1 **Reagents:** Lithium aluminum hydride
Solvents: Diethyl ether, Tetrahydrofuran; 5 min, 0 °C; 0 °C → rt; 1 h, rt
- 1.2 **Reagents:** Water
Solvents: Diethyl ether; 0 °C
- 1.3 **Reagents:** Sodium sulfate; 5 min, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


[Suppliers \(61\)](#)
[Suppliers \(51\)](#)
[Suppliers \(17\)](#)

31-520-CAS-14208120

Steps: 1 Yield: 100%

Reductive monoalkylation of aromatic and aliphatic nitro compounds and the corresponding amines with nitriles

By: Nacario, Ruel; et al

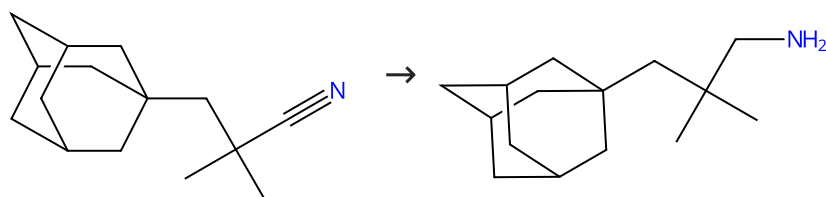
Organic Letters (2005), 7(3), 471-474.

- 1.1 **Reagents:** Ammonium formate
Catalysts: Palladium
Solvents: Methanol, Water; rt
- 1.2 4.0 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

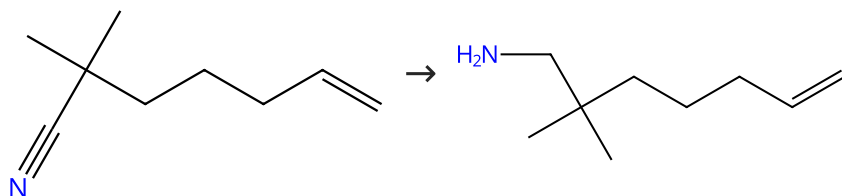
Steps: 1 Yield: 100%


[Suppliers \(5\)](#)
[Suppliers \(5\)](#)

31-520-CAS-19908009	Steps: 1 Yield: 100%	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt		By: Zhang, Hongwei; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols		

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



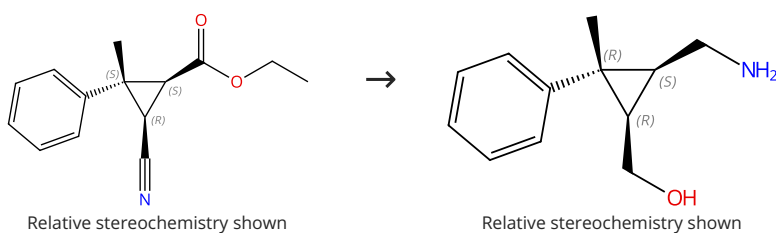
Suppliers (5)

Suppliers (15)

31-520-CAS-19908010	Steps: 1 Yield: 100%	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt		By: Zhang, Hongwei; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols		

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



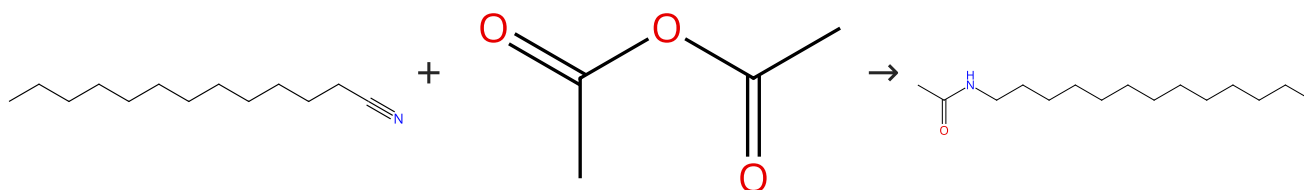
Relative stereochemistry shown

Relative stereochemistry shown

31-614-CAS-37133672	Steps: 1 Yield: 100%	Highly Diastereoselective Synthesis of Polysubstituted Cyclopropanecarbonitriles via Palladium-Catalyzed Cyanoesterification of Cyclopropenes
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 0 °C; 0 °C → 50 °C; 2 h, 50 °C; 50 °C → rt; 12 h, rt		By: Li, Can; et al
1.2 Reagents: Sodium hydroxide Solvents: Water; 0 °C; 30 min, 0 °C		Organic Letters (2023), 25(27), 5128-5133.
Experimental Protocols		

Scheme 6 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (47)

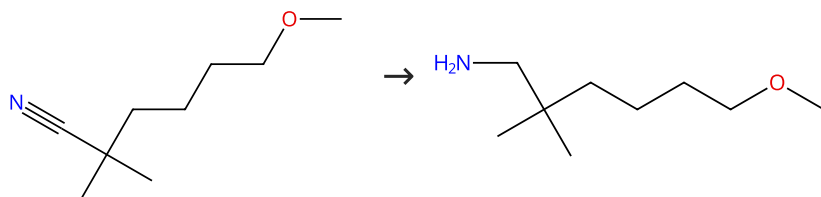
Suppliers (84)

Supplier (1)

31-520-CAS-8624587	Steps: 1 Yield: 100%	Hydrogenation of nitriles to primary amines
1.1 Reagents: Sodium acetate, Hydrogen Catalysts: Nickel Solvents: Acetic anhydride		By: Gould, Francis E.; et al Journal of Organic Chemistry (1960), 25, 1658-60.
31-520-CAS-12886184	Steps: 1 Yield: 100%	Hydrogenation of nitriles to primary amines
1.1 Reagents: Sodium acetate, Hydrogen Catalysts: Nickel		By: Gould, Francis E.; et al Journal of Organic Chemistry (1960), 25, 1658-60.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



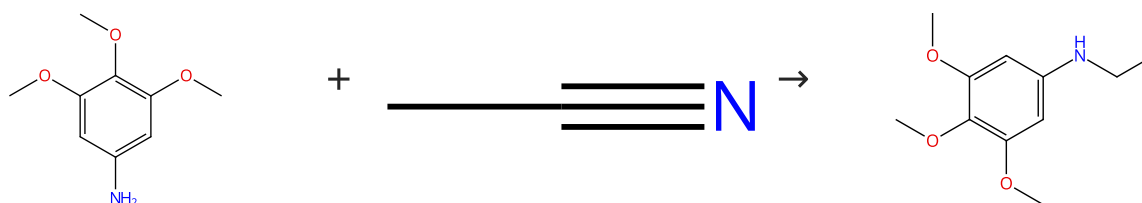
Suppliers (3)

Suppliers (3)

31-520-CAS-19908008	Steps: 1 Yield: 100%	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt		By: Zhang, Hongwei; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

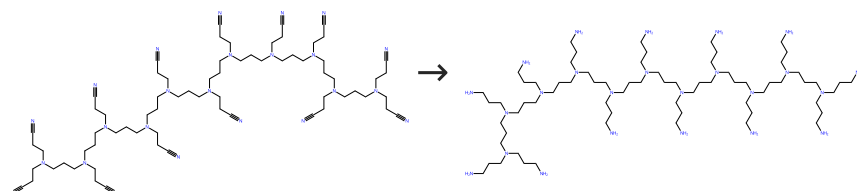
Suppliers (428)

Suppliers (18)

31-317-CAS-8614277	Steps: 1 Yield: 100%	Reductive and catalytic monoalkylation of primary amines using nitriles as an alkylating reagent
1.1 Reagents: Hydrogen Catalysts: Palladium Solvents: Methanol; 22 h, 20 °C		By: Sajiki, Hironao; et al
Experimental Protocols		Organic Letters (2004), 6(26), 4977-4980.

Scheme 9 (1 Reaction)

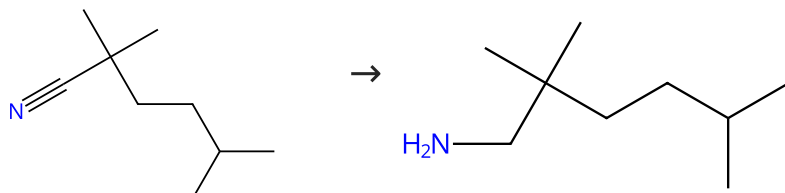
Steps: 1 Yield: 100%



31-520-CAS-12204509 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydroxide, Hydrogen Catalysts: Nickel Solvents: Methanol, Water; 1 - 4 d, 50 bar, rt	Parameters influencing the release of tertiary alcohols from the surface of "spherical" dendrimers and "linear" stylomers by neighbouring-group-assisted hydrolysis of 2- carbamoyl benzoates By: Trachsel, Alain; et al Chemistry - A European Journal (2009), 15(12), 2846-2860.
Experimental Protocols	

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



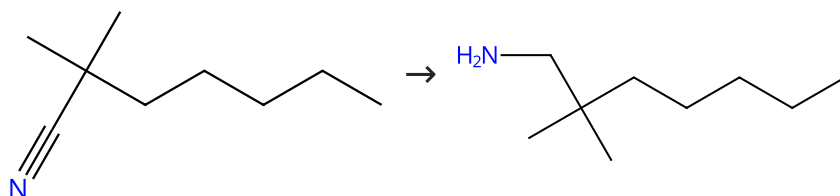
Suppliers (7)

Suppliers (8)

31-520-CAS-21017921 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt 1.2 Reagents: Sodium hydroxide Solvents: Water	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds By: Zhang, Hongwei; et al Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols	

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



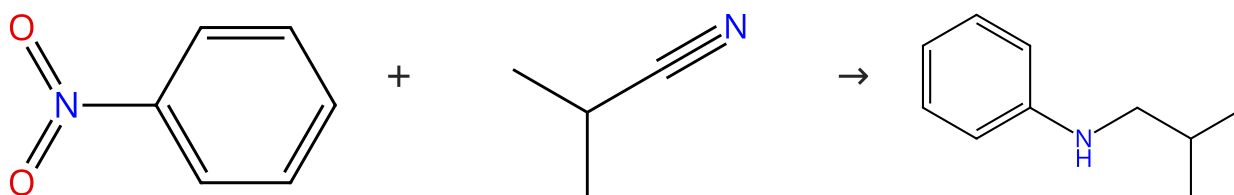
Suppliers (8)

Suppliers (33)

31-520-CAS-19908006 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt 1.2 Reagents: Sodium hydroxide Solvents: Water	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds By: Zhang, Hongwei; et al Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols	

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (103)

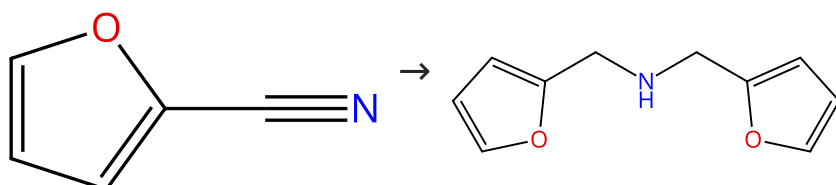
Suppliers (45)

Suppliers (21)

31-520-CAS-8281282	Steps: 1 Yield: 100%	Reductive and catalytic monoalkylation of primary amines using nitriles as an alkylating reagent
1.1 Reagents: Hydrogen Catalysts: Palladium Solvents: Methanol; 19 h, 20 °C		By: Sajiki, Hironao; et al
Experimental Protocols		Organic Letters (2004), 6(26), 4977-4980.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



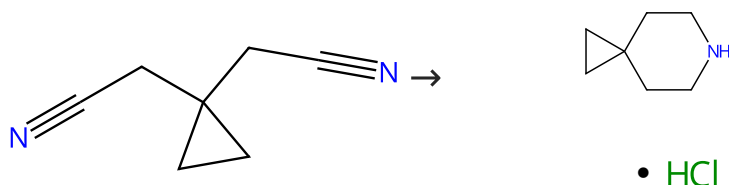
Suppliers (81)

Suppliers (48)

31-520-CAS-23896368	Steps: 1 Yield: 100%	Quasi Pd₁Ni single-atom surface alloy catalyst enables hydrogenation of nitriles to secondary amines
1.1 Reagents: Hydrogen Catalysts: Nickel alloy, base, Ni 50,Pd 50 Solvents: Ethanol; 2 h, 0.6 M Pa, 80 °C		By: Wang, Hengwei; et al
Experimental Protocols		Nature Communications (2019), 10(1), 1-9.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



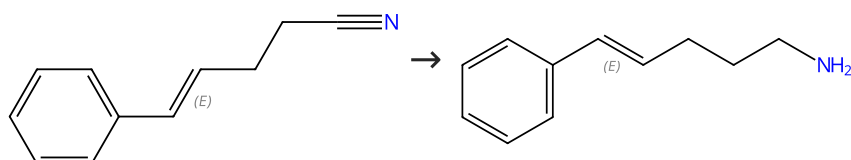
Suppliers (51)

Suppliers (64)

31-520-CAS-7009392	Steps: 1 Yield: 100%	Straightforward access to cyclic amines by dinitriles reduction
1.1 Reagents: Poly(methylhydrosiloxane) Catalysts: Titanium isopropoxide Solvents: Toluene; 24 h, 60 °C; 60 °C → rt		By: Laval, Stephane; et al
1.2 Reagents: Hydrochloric acid Solvents: Water; acidified		Tetrahedron (2014), 70(4), 975-983.
Experimental Protocols		

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

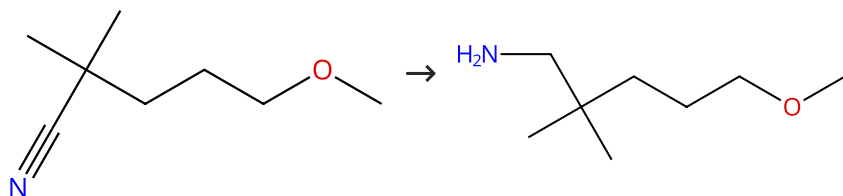
Suppliers (7)

Suppliers (9)

31-520-CAS-23751039	Steps: 1 Yield: 100%	Copper-catalyzed Beckmann-type fragmentation of less-strained cycloketoxime esters
1.1 Reagents: Lithium aluminum hydride Solvents: Diethyl ether; 0 °C; 2 h, 0 °C → reflux		By: Zhao, Binlin; et al
Experimental Protocols		Organic Chemistry Frontiers (2021), 8(12), 2985-2989.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



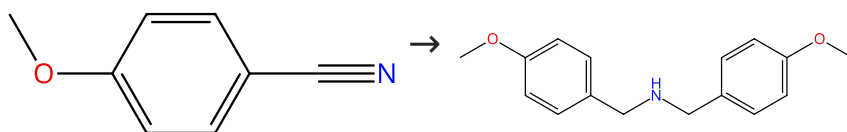
Suppliers (2)

Suppliers (8)

31-520-CAS-21017922	Steps: 1 Yield: 100%	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt		By: Zhang, Hongwei; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols		

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



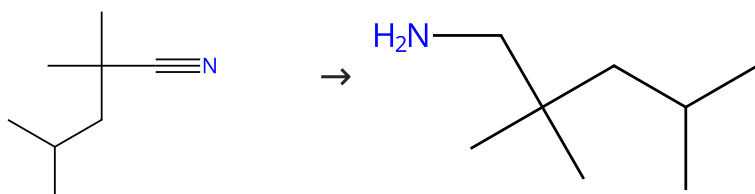
Suppliers (88)

Suppliers (70)

31-520-CAS-23896647	Steps: 1 Yield: 100%	Quasi Pd₁Ni single-atom surface alloy catalyst enables hydrogenation of nitriles to secondary amines
1.1 Reagents: Hydrogen Catalysts: Nickel alloy, base, Ni 50, Pd 50 Solvents: Ethanol; 2 h, 0.6 MPa, 80 °C		By: Wang, Hengwei; et al
Experimental Protocols		Nature Communications (2019), 10(1), 1-9.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



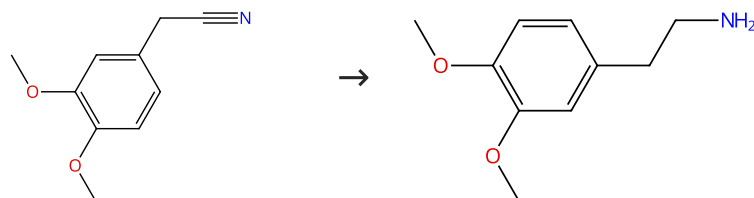
Suppliers (36)

Suppliers (13)

31-520-CAS-19908007	Steps: 1 Yield: 100%	Copper-Catalyzed Amide Radical-Directed Cyanation of Unactivated C_{sp}³-H Bonds
1.1 Reagents: Lithium aluminum hydride Solvents: Tetrahydrofuran; 12 h, rt		By: Zhang, Hongwei; et al
1.2 Reagents: Sodium hydroxide Solvents: Water		Organic Letters (2019), 21(6), 1921-1925.
Experimental Protocols		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



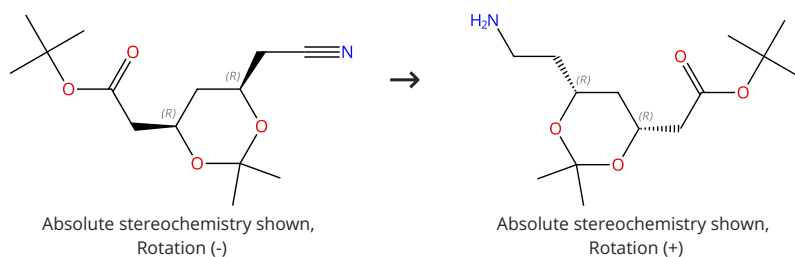
Suppliers (103)

Suppliers (64)

31-520-CAS-6346361	Steps: 1 Yield: 100%	Trapping of iminiums by the indole nucleus during catalytic hydrogenation of nitriles: a rapid synthesis of tetrahydro-β-carbolines
1.1 Reagents: Hydrogen Catalysts: Palladium Solvents: Acetic acid		By: Diker, Khalid; et al
		Tetrahedron Letters (1995), 36(14), 2497-500.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (+)

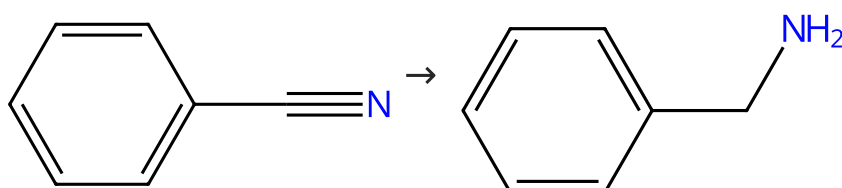
Suppliers (69)

Suppliers (72)

31-614-CAS-39827446	Steps: 1 Yield: 100%	An efficient and stable LDH-derived Ni catalyst for highly selective hydrogenation of nitriles to primary amines
1.1 Reagents: Hydrogen Catalysts: Aluminum alloy, nonbase, Al, Ni, Zr Solvents: Methanol, Ammonia; 1 MPa, 80 °C		By: Xu, Yanlin; et al
Experimental Protocols		Chemical Engineering Science (2024), 292, 120004.

Scheme 21 (14 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (77)

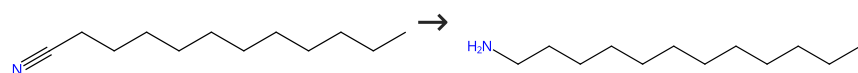
Suppliers (81)

31-520-CAS-14813378 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, lithium (1:1), (T-4)- Solvents: Tetrahydrofuran; 12 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Reaction of lithium cyanoaluminum hydride with selected organic compounds containing representative functional groups. Comparison of reducing characteristics between lithium and sodium cyanoaluminum hydrides By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2009), 30(7), 1588-1592.
31-520-CAS-16881782 Steps: 1 Yield: 100% 1.1 Reagents: (T-4)-[N,2-Dimethyl-N-(1-methylethyl)-2-propanamine]trihydroboron Solvents: Tetrahydrofuran, 1,4-Dioxane; 6 h, reflux 1.2 Reagents: Hydrochloric acid Solvents: Water; 1 h, reflux; cooled 1.3 Reagents: Sodium hydroxide Solvents: Water; neutralized, cooled	Molecular Addition Compounds. 15. Synthesis, Hydroboration, and Reduction Studies of New, Highly Reactive tert-Butyldialkylamine-Borane Adducts By: Brown, Herbert C.; et al Journal of Organic Chemistry (1999), 64(17), 6263-6274.
31-520-CAS-9139880 Steps: 1 Yield: 100% 1.1 Reagents: (T-4)-[N-Ethyl-N-(1-methylethyl)benzenamine] trihydroboron Solvents: Tetrahydrofuran; 6 h, reflux; cooled 1.2 Reagents: Hydrochloric acid, Water; 1 h, reflux; cooled 1.3 Reagents: Sodium hydroxide Solvents: Water; neutralized Experimental Protocols	Molecular Addition Compounds. 11. N-Ethyl-N-isopropyl aniline-Borane, A Superior Reagent for Hydroborations and Reductions By: Brown, Herbert C.; et al Journal of Organic Chemistry (1998), 63(15), 5154-5163.
31-614-CAS-36511746 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonia Catalysts: Nickel, Rhenium, Silica Solvents: Methanol, Water; 90 min, 1 M Pa, 120 °C Experimental Protocols	Highly efficient and stable rhenium modified nickel catalyst for hydrogenation of nitriles to primary amines By: Ma, Zhuo; et al Chemical Engineering Journal (Amsterdam, Netherlands) (2023), 466, 143238.
31-614-CAS-35932800 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonium hydroxide Catalysts: Nickel, Aluminum zinc oxide Solvents: Hexane; 14 h, 10 bar, 25 °C Experimental Protocols	Room-temperature synthesis of primary amines by selective hydrogenation of nitriles over ZnAlO_x supported Ni catalysts By: Zhao, Huanyu; et al Applied Catalysis, A: General (2023), 655, 119115.
31-520-CAS-22585687 Steps: 1 Yield: 99% 1.1 Reagents: Phenylsilane Catalysts: Manganese, bromotricarbonyl[2-(6-methyl-2-pyridinyl-κN)-1H-benzimidazole-κN³]-, (OC-6-44)- Solvents: Tetrahydrofuran; 24 h, 120 °C; cooled 1.2 Reagents: Sodium hydroxide Solvents: Methanol, Water; cooled; 6 h, rt	Benzimidazole fragment containing Mn-complex catalyzed hydrosilylation of ketones and nitriles By: Ganguli, Kasturi; et al Tetrahedron (2020), 76(37), 131439.
31-520-CAS-19857168 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonia Catalysts: Nickel Solvents: Methanol, Water; 6 h, 2.5 bar, 80 °C	Nitrogen-Doped Carbon-Supported Nickel Nanoparticles: A Robust Catalyst to Bridge the Hydrogenation of Nitriles and the Reductive Amination of Carbonyl Compounds for the Synthesis of Primary Amines By: Zhang, Yangmin; et al ChemSusChem (2019), 12(6), 1246-1255.

31-520-CAS-19623659 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen Catalysts: 4-Nitro-1,2-phenylenediamine (complexes with Pd) Solvents: Water; 3 h, pH 1, rt 1.2 Reagents: Sodium bicarbonate Solvents: Water; basified, rt	Mild palladium-catalysed highly efficient hydrogenation of CN, C-NO₂, and C=O bonds using H₂ of 1 atm in H₂O By: Liu, Yaxu; et al Green Chemistry (2019), 21(4), 830-838.
31-520-CAS-1113604 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen Catalysts: Ruthenium, [(1,2,5,6-η)-1,5-cyclooctadiene]bis[(1,2,3-η)-2-methyl-2-propenyl]-, Phosphine, bis[2-[[bis(1,1-dimethylethyl)phosphino]methyl]phenyl]phenyl- Solvents: Isopropanol; 15 bar, rt; 17 h, rt → 50 °C 1.2 Reagents: Hydrochloric acid Solvents: Methanol; 30 min, rt Experimental Protocols	A Mild and Base-Free Protocol for the Ruthenium-Catalyzed Hydrogenation of Aliphatic and Aromatic Nitriles with Tridentate Phosphine Ligands By: Adam, Rosa; et al ChemCatChem (2016), 8(7), 1329-1334.
31-520-CAS-14964485 Steps: 1 Yield: 99% 1.1 Reagents: Sodium triethylborohydride Catalysts: (7-4)-[2-[Bis(1-methylethyl)phosphino-κP]-N-[[3-[bis(1-methylethyl)phosphino]phenyl]methyl]benzenemet hanamine-κM]dibromoiron Solvents: Tetrahydrofuran; rt; 20 min, rt 1.2 Reagents: Hydrogen, Potassium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 16 h, 60 bar, 140 °C Experimental Protocols	Selective hydrogenation of nitriles to primary amines catalyzed by a novel iron complex By: Chakraborty, Subrata; et al Chemical Communications (Cambridge, United Kingdom) (2016), 52(9), 1812-1815.
31-520-CAS-10780198 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen Catalysts: Alumina, Potassium, Ruthenium, Triruthenium dodecacarbonyl Solvents: Nonane, 1,4-Dioxane; 82 h, 101.3 kPa, 343 K	Formation and nitrile hydrogenation performance of Ru nanoparticles on a K-doped Al₂O₃ surface By: Muratsugu, Satoshi; et al Physical Chemistry Chemical Physics (2015), 17(38), 24791-24802.
31-520-CAS-11037473 Steps: 1 Yield: 99% 1.1 Catalysts: Palladium alloy, base, Pd 81, Ni 19 (graphene-supported) Solvents: Methanol, Water; 5 min, rt 1.2 Reagents: Ammonia borane; 10 min, rt Experimental Protocols	Tandem Dehydrogenation of Ammonia Borane and Hydrogenation of Nitro/Nitrile Compounds Catalyzed by Graphene-Supported NiPd Alloy Nanoparticles By: Goksu, Haydar; et al ACS Catalysis (2014), 4(6), 1777-1782.
31-520-CAS-7879990 Steps: 1 Yield: 99% 1.1 Catalysts: Potassium <i>tert</i>-butoxide, 1,1-Bis(diphenylphosphino)ferrocene, Ruthenium, [(1,2,5,6-η)-1,5-cyclooctadiene]bis[(1,2,3-η)-2-methyl-2-propenyl]- Solvents: Toluene; 10 min, rt 1.2 Reagents: Hydrogen; 1 h, 50 bar, 80 °C Experimental Protocols	A general and environmentally benign catalytic reduction of nitriles to primary amines By: Enthaler, Stephan; et al Chemistry - A European Journal (2008), 14(31), 9491-9494.
31-520-CAS-10364410 Steps: 1 Yield: 99% 1.1 Reagents: Phosphoric acid Catalysts: Samarium iodide (SmI₂) Solvents: Tetrahydrofuran	Reductions of carboxylic acids and their derivatives using a samarium diiodide-acid system By: Kamochi, Yasuko; et al Tetrahedron (1992), 48(21), 4301-12.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

Suppliers (83)

31-520-CAS-11615974

Steps: 1 Yield: 100%

Sodium reduction of fatty acid esters

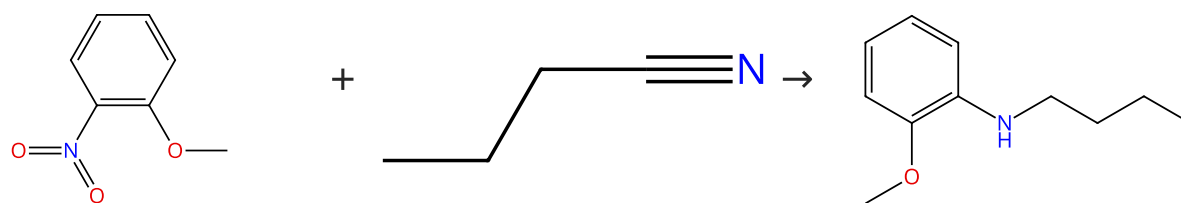
1.1 Reagents: Sodium
Solvents: Xylene

By: Hansley, V. L.

Industrial and Engineering Chemistry (1947), 39, 55-62.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (54)

Suppliers (51)

Suppliers (24)

31-520-CAS-12903042

Steps: 1 Yield: 100%

Reductive monoalkylation of aromatic and aliphatic nitro compounds and the corresponding amines with nitriles

1.1 Reagents: Ammonium formate
Catalysts: Palladium
Solvents: Methanol, Water; rt

By: Nacario, Ruel; et al

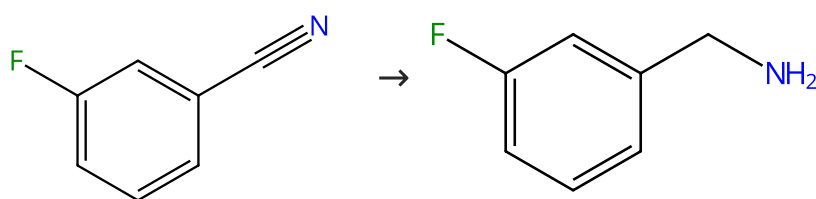
Organic Letters (2005), 7(3), 471-474.

1.2 3.1 h, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (77)

31-520-CAS-20062611

Steps: 1 Yield: 100%

Turning the product selectivity of nitrile hydrogenation from primary to secondary amines by precise modification of Pd/Si C catalysts using NiO nanodots

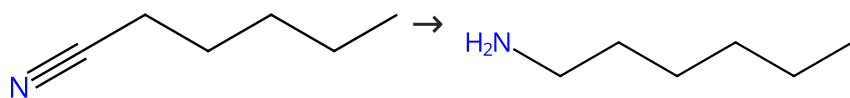
1.1 Reagents: Hydrogen
Catalysts: Palladium
Solvents: Ethanol; 2 h, 1 atm, 30 °C

By: Jiao, Zhi-Feng; et al

Catalysis Science & Technology (2019), 9(9), 2266-2272.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (33)

Suppliers (71)

31-520-CAS-2035769

Steps: 1 Yield: 100%

Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.

1.1 **Reagents:** Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)-

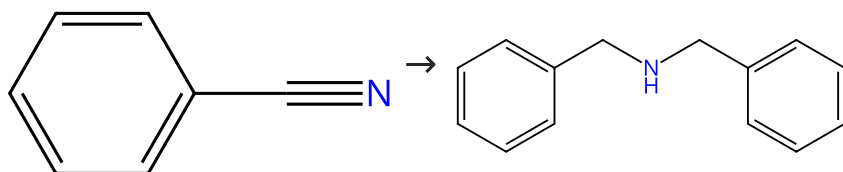
Solvents: Tetrahydrofuran; 72 h, 25 °C

1.2 **Reagents:** Hydrochloric acid

Solvents: Water

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (91)

31-520-CAS-23896813

Steps: 1 Yield: 100%

Quasi Pd₁Ni single-atom surface alloy catalyst enables hydrogenation of nitriles to secondary amines

By: Wang, Hengwei; et al

Nature Communications (2019), 10(1), 1-9.

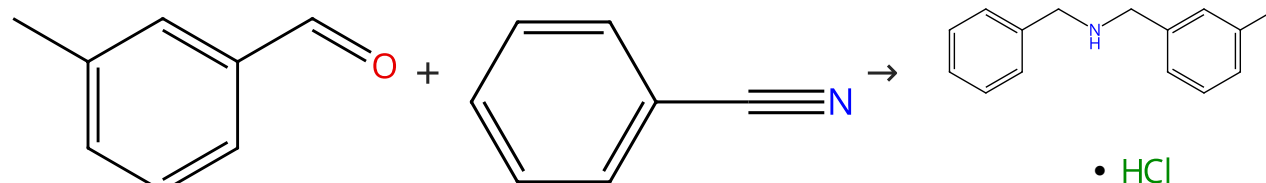
1.1 **Reagents:** Hydrogen

Catalysts: Nickel alloy, base, Ni 50, Pd 50**Solvents:** Ethanol; 2 h, 0.6 MPa, 80 °C

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (87)

Suppliers (77)

Suppliers (24)

31-520-CAS-23026975

Steps: 1 Yield: 99%

General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones

By: Schoenauer, Timon; et al

Chemistry - A European Journal (2021), 27(5), 1609-1614.

1.1 **Reagents:** Hydrogen

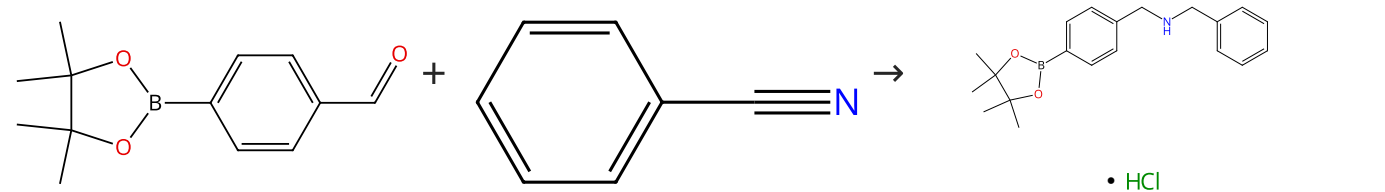
Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported)**Solvents:** 2-Methyltetrahydrofuran; 20 h, 1.5 MPa, 100 °C; 100 °C → rt

1.2 **Reagents:** Hydrochloric acid

Solvents: Diethyl ether; rt

Experimental Protocols

Scheme 28 (1 Reaction)



Suppliers (84)

Suppliers (77)

31-520-CAS-23033369

Steps: 1 Yield: 99%

General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones

By: Schoenauer, Timon; et al

Chemistry - A European Journal (2021), 27(5), 1609-1614.

1.1 Reagents: Hydrogen

Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported)

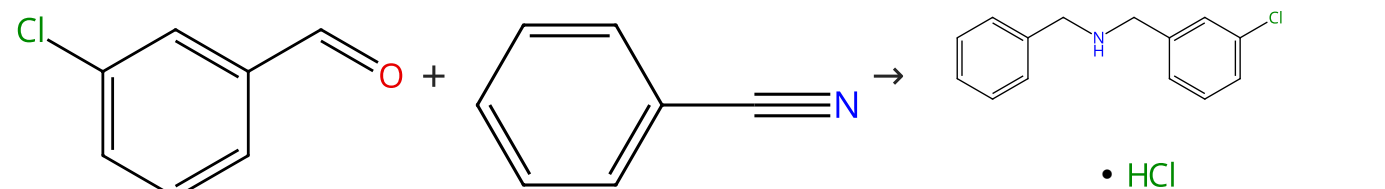
Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 MPa, 100 °C; 100 °C → rt

1.2 Reagents: Hydrochloric acid

Solvents: Diethyl ether; rt

Experimental Protocols

Scheme 29 (1 Reaction)



Suppliers (88)

Suppliers (77)

Suppliers (2)

31-520-CAS-23033058

Steps: 1 Yield: 99%

General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones

By: Schoenauer, Timon; et al

Chemistry - A European Journal (2021), 27(5), 1609-1614.

1.1 Reagents: Hydrogen

Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported)

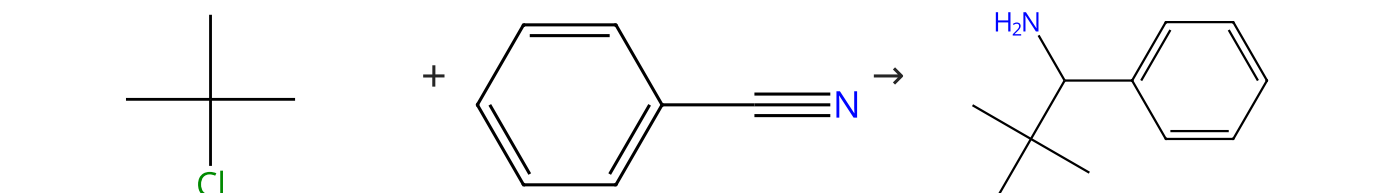
Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 MPa, 100 °C; 100 °C → rt

1.2 Reagents: Hydrochloric acid

Solvents: Diethyl ether; rt

Experimental Protocols

Scheme 30 (1 Reaction)



Suppliers (60)

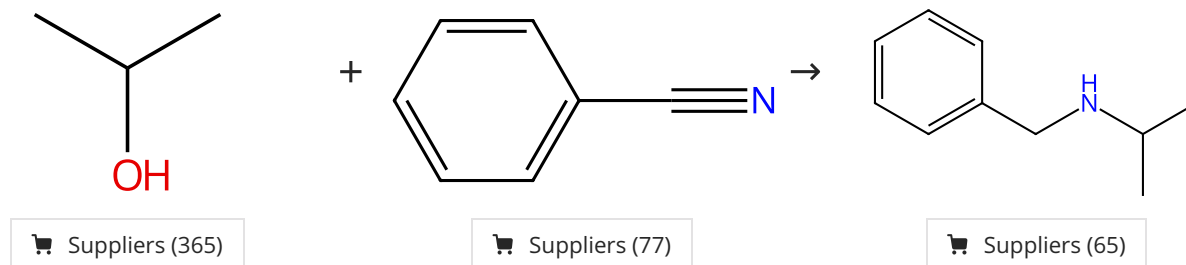
Suppliers (77)

Suppliers (64)

31-520-CAS-14078553	Steps: 1 Yield: 99%	Tandem alkylation-reduction of nitriles. Synthesis of branched primary amines By: Weiberth, Franz J.; et al Journal of Organic Chemistry (1986), 51(26), 5338-41.
1.1 Reagents: Magnesium Catalysts: Copper bromide (CuBr) Solvents: Tetrahydrofuran		
1.2 Reagents: Lithium Solvents: Ammonia		

Scheme 31 (1 Reaction)

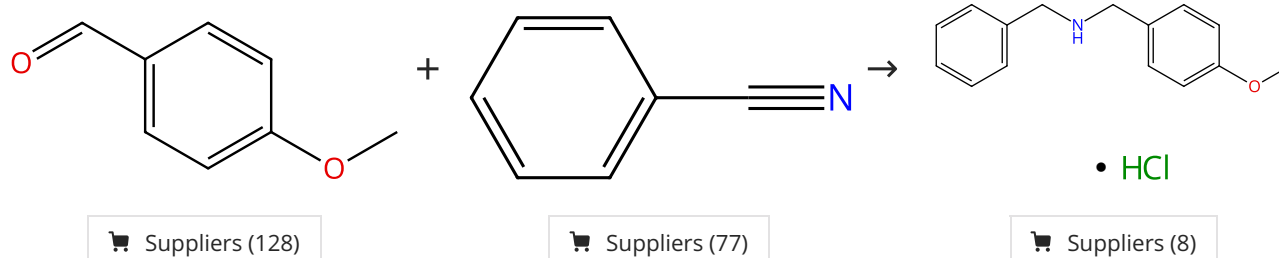
Steps: 1 Yield: 99%



31-520-CAS-2072417	Steps: 1 Yield: 99%	Ruthenium-catalyzed transfer hydrogenation of nitriles: reduction and subsequent N-monoalkylation to secondary amines By: Werkmeister, Svenja; et al European Journal of Organic Chemistry (2013), 2013(18), 3671-3674.
1.1 Catalysts: Dichlorotris(triphenylphosphine)ruthenium; 5 min, rt		
1.2 Reagents: Sodium hydroxide; 5 min, rt		
1.3 6 h, 120 °C		
Experimental Protocols		

Scheme 32 (1 Reaction)

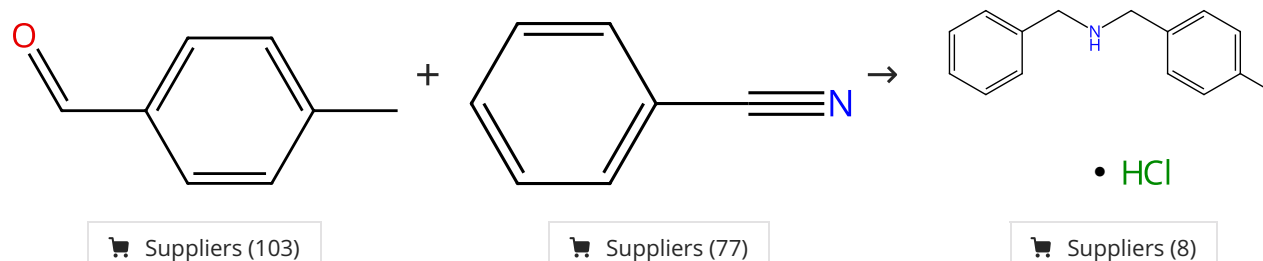
Steps: 1 Yield: 99%



31-520-CAS-23033878	Steps: 1 Yield: 99%	General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones By: Schoenauer, Timon; et al Chemistry - A European Journal (2021), 27(5), 1609-1614.
1.1 Reagents: Hydrogen Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported) Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 M Pa, 100 °C; 100 °C → rt		
1.2 Reagents: Hydrochloric acid Solvents: Diethyl ether; rt		
Experimental Protocols		

Scheme 33 (1 Reaction)

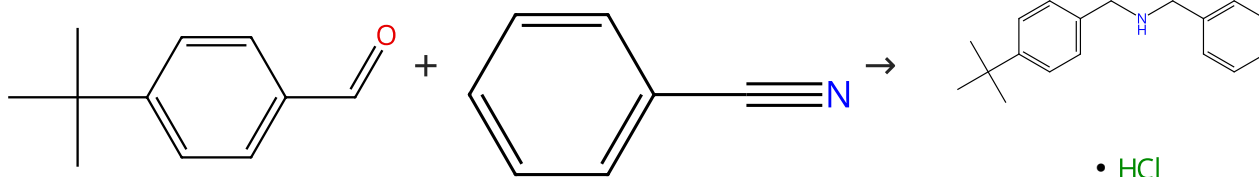
Steps: 1 Yield: 99%



31-520-CAS-23028817 Steps: 1 Yield: 99%	General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones
1.1 Reagents: Hydrogen Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported) Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 M Pa, 100 °C; 100 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Diethyl ether; rt	By: Schoenauer, Timon; et al Chemistry - A European Journal (2021), 27(5), 1609-1614.
Experimental Protocols	

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



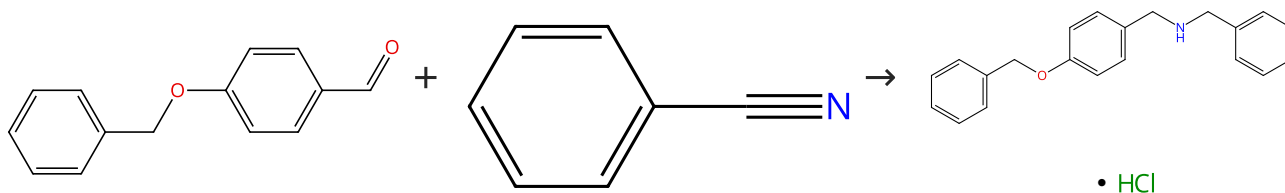
Suppliers (75)

Suppliers (77)

31-520-CAS-23032929 Steps: 1 Yield: 99%	General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones
1.1 Reagents: Hydrogen Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported) Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 M Pa, 100 °C; 100 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Diethyl ether; rt	By: Schoenauer, Timon; et al Chemistry - A European Journal (2021), 27(5), 1609-1614.
Experimental Protocols	

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

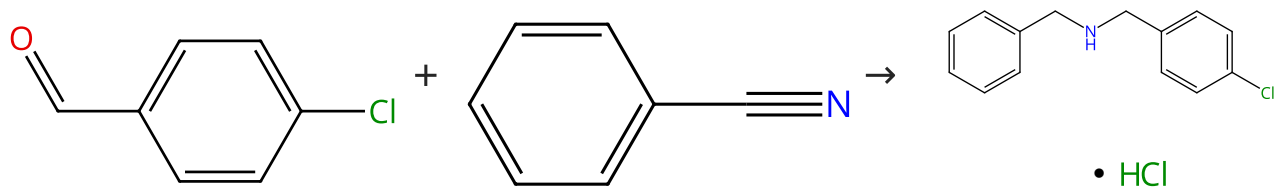
Suppliers (77)

Supplier (1)

31-520-CAS-23027126 Steps: 1 Yield: 99%	General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones
1.1 Reagents: Hydrogen Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported) Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 M Pa, 100 °C; 100 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Diethyl ether; rt	By: Schoenauer, Timon; et al Chemistry - A European Journal (2021), 27(5), 1609-1614.
Experimental Protocols	

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (77)

Suppliers (2)

31-520-CAS-23033493

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Hydrogen
Catalysts: Silicon monocarbide, Cobalt (nitrogen-doped silicon carbide supported)
Solvents: 2-Methyltetrahydrofuran; 20 h, 1.5 MPa, 100 °C; 100 °C → rt
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Diethyl ether; rt

Experimental Protocols

General Synthesis of Secondary Alkylamines by Reductive Alkylation of Nitriles by Aldehydes and Ketones

By: Schoenauer, Timon; et al

Chemistry - A European Journal (2021), 27(5), 1609-1614.